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PSYC4411

11:21 AM

PREDICTION WITH ACCOUNTABILITY

CLASSIFICATION, UNCERTAINTY, AND EVALUATION

Week 5 · Semester 1, 2026

TODAY'S PLAN

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1. From regression to classification — the conceptual shift
2. Logistic regression — probabilities, not just labels
3. The confusion matrix — four types of prediction
4. Classification metrics — beyond accuracy
5. Decision thresholds — 0.5 is a choice, not a law
6. Decision trees and random forests — interpretable models
7. Fairness and ethics — when algorithms affect people
8. Getting ready for Week 6

FROM REGRESSION TO CLASSIFICATION

Numbers to categories

THE SHIFT: NUMBERS → CATEGORIES

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REGRESSION

Predict a **number**

"Depression score = 14.3"

Metric: How far off? (MAE, R^2)

→

CLASSIFICATION

Predict a **category**

"Elevated depression: Yes/No"

Metric: Which errors? (more complex)

When you classify people, the **type** of mistake matters as much as the **number** of mistakes.

BINARY CLASSIFICATION

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- Target has exactly **two values** — typically coded as 0 and 1
- In psychology, this is everywhere:
  - Clinical diagnosis: yes / no
  - Treatment response: responder / non-responder
  - Risk assessment: at-risk / low-risk
  - Dropout prediction: stays / leaves
- You already know this from **logistic regression** in statistics
  - The difference: ML focuses on *prediction accuracy*, not *significance testing*

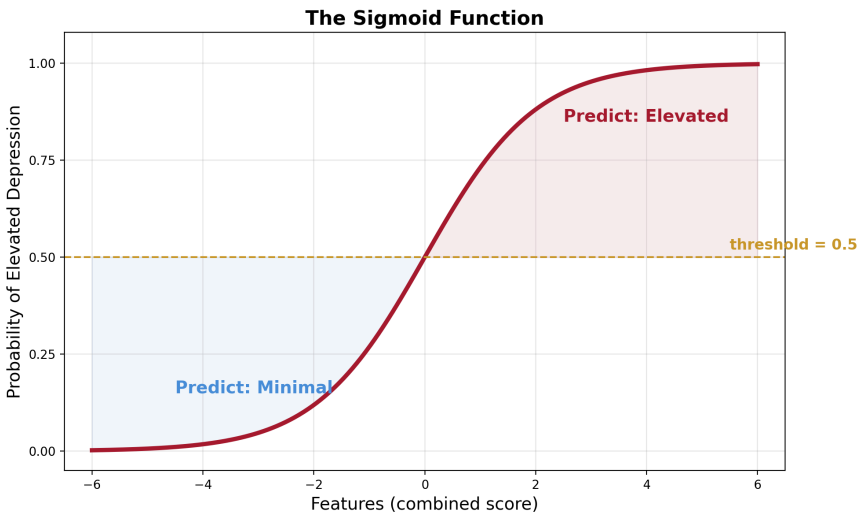
Week 6 target: Classify participants as having **elevated depression** (PHQ-9 ≥ 5) vs **minimal symptoms** (PHQ-9 < 5) using real COVID-era survey data.

LOGISTIC REGRESSION

Probabilities, not just labels

THE SIGMOID FUNCTION

Linear regression draws a **straight line**. Logistic regression fits an **S-shaped curve**.



The curve squashes any input to a value between 0 and 1 — we interpret this as a **probability**.

PROBABILITY, THEN DECISION

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- The model outputs: **“73% probability of elevated depression”**
  - Not just “elevated” or “not elevated”
- We choose a **decision boundary** — typically 0.5
  - Above 0.5 → classify as “elevated”
  - Below 0.5 → classify as “minimal”
- This threshold is a **choice**, not a law — more on this soon

> The probability output is powerful: it lets us adjust **how cautious** the model is.

STATS CONNECTION

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Logistic regression has been a **workhorse of behavioural research** for decades.

IN STATISTICS

- "Is this predictor **significant**?"
- Focus on p-values, odds ratios
- Explain relationships

IN ML

- "Can the model **correctly identify** who belongs to each group?"
- Focus on accuracy, precision, recall
- Predict new cases

Same model, different goals. ML asks: "How well does it work on *new* data?"

THE CONFUSION MATRIX

Four types of prediction

THE CONFUSION MATRIX

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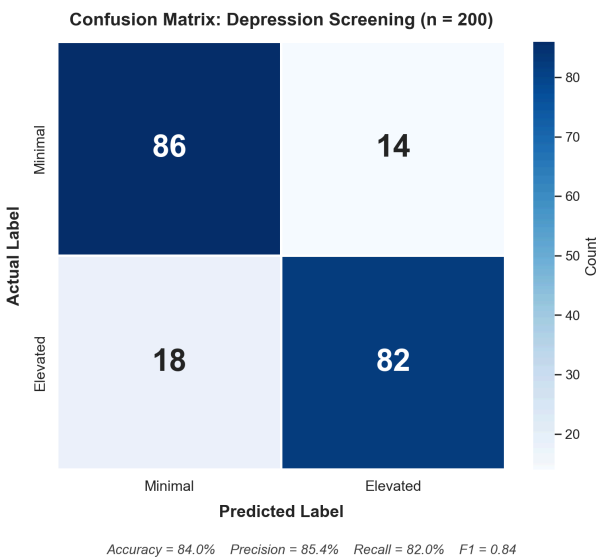
	ActuallyElevated	ActuallyMinimal
PredictedElevated	**True Positive** Has depression, model catches it → referred for support	**False Positive** Actually fine, model flags them → unnecessary follow-up
PredictedMinimal	**False Negative** Has depression, model misses it → no support offered	**True Negative** Actually fine, model agrees → correctly ruled out

In screening, **false negatives** are usually the more dangerous error — we miss someone who needs help.

WHAT THIS LOOKS LIKE IN CODE

Here's a real confusion matrix from Python — this is what you'll produce in Week 6:



- **Top-left (86):** Correctly identified as minimal (True Negative)
- **Top-right (14):** Flagged as elevated but actually minimal (False Positive)
- **Bottom-left (18):** Missed — actually elevated but predicted minimal (False Negative)
- **Bottom-right (82):** Correctly identified as elevated (True Positive)

18 missed cases — people with elevated depression that the model didn't catch. Are you comfortable with that?

THINK ABOUT IT

Imagine two depression screening tools for a university:

Tool A: Catches 95% of students with elevated depression, but flags 40% of students who are fine.

Tool B: Catches 70% of students with elevated depression, but rarely flags anyone incorrectly.

Which would you choose for a walk-in counselling service?
What about for emergency crisis detection?

CLASSIFICATION METRICS

Beyond accuracy

THE ACCURACY TRAP

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Accuracy = proportion of predictions that were correct. Seems like the obvious metric.

The trap: Imagine 95% of your sample does *not* have elevated depression. A model that predicts “not elevated” for **everyone** achieves 95% accuracy — without learning anything.

- This is the **majority-class baseline** — always guess the more common class
- Any model must beat this to prove it's actually learning
- Our Week 6 data is nearly balanced (54.5% elevated) — so baseline accuracy ≈ 54.5%

Class imbalance makes accuracy even more misleading — when classes are skewed, always report F1 or AUC alongside accuracy.

THE CORE METRICS

Precision

Of everyone flagged as elevated, how many actually were?
High precision = few false alarms

Recall (Sensitivity)

Of everyone who actually had elevated depression, how many did we catch?
High recall = few missed cases

F1 Score

Harmonic mean of precision and recall — a single number that balances both.

Specificity

Of everyone who was actually minimal, how many did we correctly identify?
The mirror image of recall.

THE PRECISION–RECALL TRADE-OFF

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HIGH PRECISION

When you flag someone, you're almost always right

But you might miss people who need help

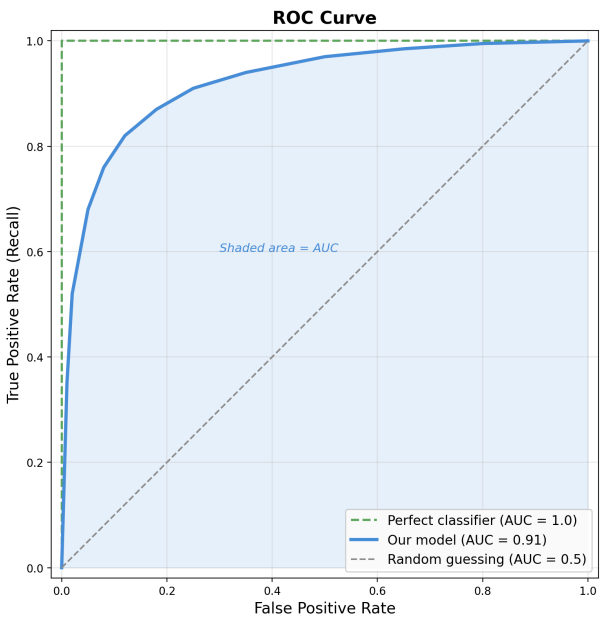
HIGH RECALL

You catch almost everyone who needs help

But you also flag many who are fine

You can't maximise both at once — making the model more aggressive at catching cases inevitably increases false alarms.

ROC CURVE AND AUC



- **ROC curve** = recall vs false positive rate at *every* threshold
- **AUC** = area under that curve
  - Perfect model: AUC = 1.0 (green dashed)
  - Random guessing: AUC = 0.5 (grey dashed)
- Evaluates the model **across all thresholds** at once
- **Interpretation:** "If I pick one elevated and one minimal person at random, how often does the model rank the elevated person higher?"

DECISION THRESHOLDS

0.5 is a choice, not a law

THE THRESHOLD IS A CHOICE

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Most classifiers output a **probability**. The threshold for converting that to a label is **your decision**.

Low (0.3)

Cast a **wider net**
More true positives
More false alarms

Default (0.5)

Balanced approach
Equal treatment
of both errors

High (0.7)

More **conservative**
Fewer false alarms
More missed cases

COSTS OF DIFFERENT ERRORS

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Screening Programme

Missing someone who needs help is **worse** than a false alarm.

→ Lower the threshold (e.g., 0.3)

→ Prioritise **recall**

Clinical Trial Selection

False positives **waste expensive resources**.

→ Raise the threshold (e.g., 0.7)

→ Prioritise **precision**

The right threshold depends on the **real-world consequences** of each type of error.

THINK ABOUT IT

You've built a classifier that predicts which first-year students are likely to **drop out** of university.

Low threshold (0.3): many students flagged, extra support spread thin.
High threshold (0.7): only the most at-risk get support, but some are missed.

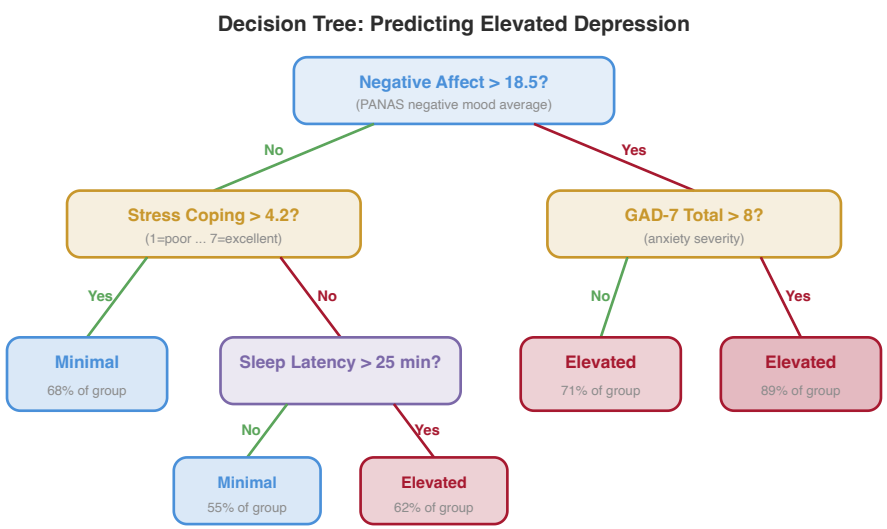
What factors should guide your choice?

DECISION TREES

If-then rules for prediction

HOW DECISION TREES WORK

A decision tree asks a series of **yes/no questions** about your features, creating a flowchart:



TREES: STRENGTHS AND WEAKNESSES

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STRENGTHS

- **Interpretable** — you can trace exactly *why* a prediction was made
- Handle non-linear relationships naturally
- No feature scaling needed
- Handle mixed data types

WEAKNESSES

- **Prone to overfitting** — unrestricted trees memorise the training data
- Sensitive to small data changes
- Greedy algorithm: locally optimal, not globally

> The downside is serious: an unrestricted tree keeps splitting until it perfectly memorises the training data. Like a clinician building an ever-more-elaborate checklist that encodes quirks of specific patients rather than general patterns.

ENSEMBLES & RANDOM FORESTS

Many trees are better than one

THE ENSEMBLE IDEA

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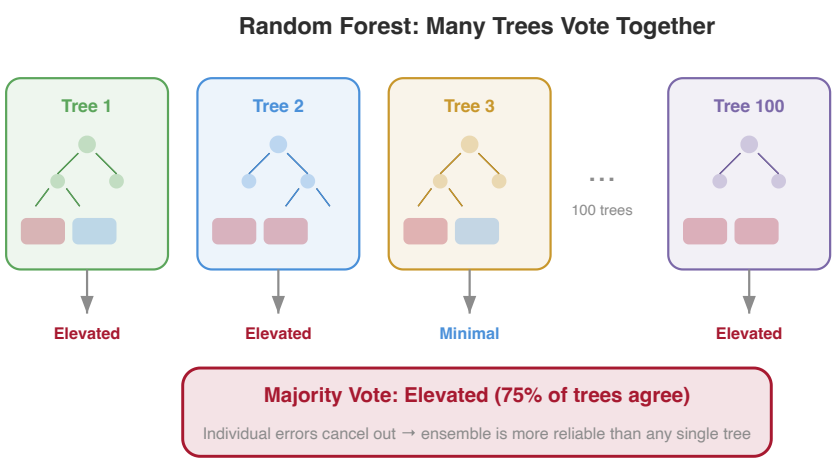
One tree is **interpretable but unstable**. What about *many* trees?

> Like asking **100 slightly different experts** for their opinion and going with the consensus.



Each tree sees a random subset of data and features — diversity reduces overfitting

WHY RANDOM FORESTS WORK

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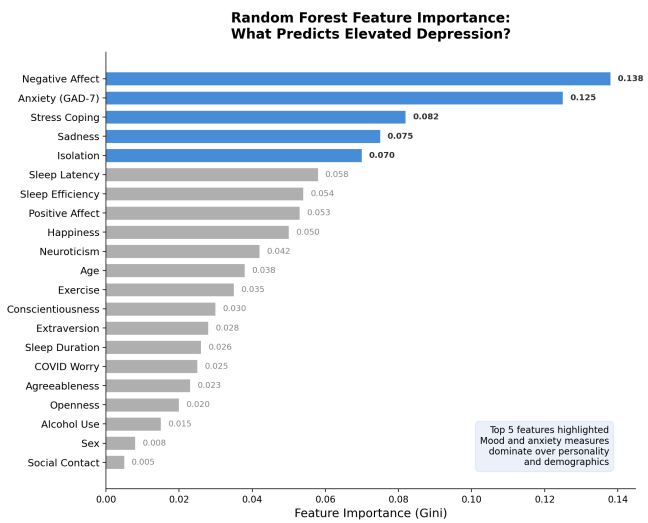
- Each tree is trained on a **random subset** of data and features
  - Different trees overfit to different quirks
- When you **average across many trees**, idiosyncratic errors cancel out
- This is the **bias–variance trade-off** from Week 3:
  - Each tree: low bias, **high variance**
  - The ensemble: keeps low bias, **dramatically reduces variance**

FEATURE IMPORTANCE

Random forests rank which features contributed most across all trees:



- **Mood and anxiety** measures dominate — negative affect, GAD-7, stress coping
- **Personality and demographics** contribute less
- This helps researchers understand *what the model relies on* — and whether those features make psychological sense

> Feature importance can also guide **feature selection** — drop low-importance features to simplify the model.

THINK ABOUT IT

Your logistic regression achieves **82% accuracy** on a depression classification task.

A decision tree achieves only **76% accuracy**.

A colleague says you should use the logistic regression because it's more accurate.

Can you think of a situation where you might prefer the **less accurate** tree?

FAIRNESS & ETHICS

When algorithms affect people

CASE STUDY: COMPAS

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COMPAS — Predicting Recidivism in Criminal Justice

An algorithm used in US courts to predict whether defendants would reoffend.

ProPublica found: Black defendants were **almost twice as likely** to be incorrectly labelled as high risk (false positives) compared to white defendants — even when controlling for prior criminal history.

- Same **overall accuracy** for both groups
- But **different error patterns** — same accuracy, different fairness
- This is a **classification problem** where the type of error matters enormously

CASE STUDY: HEALTHCARE ALGORITHM BIAS

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Obermeyer et al. (2019) — Science

A widely-used algorithm identified patients who would benefit from extra care.

The algorithm used healthcare **costs** as a proxy for health **needs**. But Black patients historically had less access to healthcare (lower costs) → the algorithm **systematically under-identified** Black patients who needed care.

The algorithm wasn't “racist” in its code — it learned from **biased data** that reflected historical inequities.

THE IMPOSSIBILITY THEOREM

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Chouldechova (2017) proved mathematically that when **base rates differ between groups**, you *cannot* simultaneously achieve:

Equal false positive rates across groups	**Equal false negative rates** across groups	**Equal predictive values** across groups

> This isn't a technical problem to solve — it's a **values question** about which type of fairness matters most in each context.

THINK ABOUT IT

You've built a depression classifier that achieves **85% accuracy overall**.

When you check across demographic groups: **90% accurate** for one group but only **72%** for another.

Should you deploy this model?
What questions would you want to ask before deciding?

COMMON MISCONCEPTIONS

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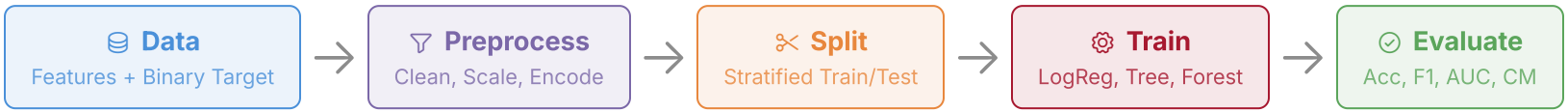
- **“Higher accuracy always means a better model”**
  - Not when classes are imbalanced. Always check precision, recall, and AUC.
- **“Random forests are always better than logistic regression”**
  - When relationships are roughly linear and data is moderate-sized, logistic regression often matches — and it's more interpretable.
- **“The model's output is the final answer”**
  - The model outputs a probability. Converting to a label requires choosing a threshold — that's a human decision.
- **“If the model is accurate, it's fair”**
  - Overall accuracy can hide very different error rates across groups.

THE CLASSIFICATION PIPELINE



Same pipeline as regression — different metrics and different model types.

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GETTING READY FOR WEEK 6

Your second challenge lab

WEEK 6: BUILD A DEFENSIBLE CLASSIFIER

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- Real data: **Boston College COVID-19 Sleep & Well-Being Study**
  - 836 participants — daily surveys + personality + demographics
- Build and compare **four classifiers**:
  - Baseline → Logistic Regression → Decision Tree → Random Forest
- Target: **Elevated depression** (PHQ-9 ≥ 5) — nearly balanced (55% / 45%)
- **Justify your threshold** — why 0.5? Why not 0.3 or 0.7?
- Report **proper classification metrics** — not just accuracy

NEW LLM SKILL: REFACTORING

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Week 2: **Prompting** · Week 4: **Debugging** · Week 6: **Refactoring**

Weak prompt

"Clean up my code."

Strong prompt

"Refactor this pipeline to: (1) separate data loading from modelling, (2) add assertions to verify data shape after each merge, (3) create a reusable function for fitting and evaluating a model, (4) add docstrings, (5) add comments explaining *why* each step is done. Prioritise readability over cleverness."

Refactoring = making code cleaner and more maintainable *without changing what it does*.

BEFORE NEXT WEEK

11:21 AM

DOWNLOAD THE DATA

```
conda activate psyc4411-env
cd weeks/week-06-lab/data
python download_data.py
```

Downloads ~22 MB of CSV files. If you've already done this for a Week 4 bonus challenge, the files will already be there.

REVIEW

- Read the companion reading if you haven't
- Review: confusion matrix, precision, recall, F1, AUC
- Know the difference between a tree and a forest

KEY REFERENCES

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- **Steyerberg et al. (2010)** — Assessing prediction model performance
  - doi.org/10.1097/EDE.0b013e3181c30fb2
- **Obermeyer et al. (2019)** — Dissecting racial bias in healthcare algorithms
  - doi.org/10.1126/science.aax2342
- **Chouldechova (2017)** — Fair prediction with disparate impact
  - doi.org/10.1089/big.2016.0047
- **James et al. (2023)** — *Intro to Statistical Learning (Python)*, Ch 4
  - statlearning.com — Free PDF

Full reading list: readings.md

QUESTIONS?

Next week: Build a Defensible Classifier — Trees, Ensembles, and Real Data

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